Exit Times and Exit Distributions for Absorbing Discrete-Time Markov Chains

Theory, Derivation and Monte Carlo Validation

Sombit Mazumder
Shaumik Khanna

STPR 2026 Project
Chennai Mathematical Institute

November 5, 2026

Outline

1. Problem Statement
2. Preliminaries: Notation and Classification of States
3. Partitioning the Transition Matrix
4. Part 1: Exit Time
5. Part 2: Exit Distribution
6. Implementation and Validation

Problem Statement

Write a Python program that accepts user-defined state classifications (transient versus recurrent/absorbing states) alongside the transition matrix, simulates a large number independent trajectories (given a user-selected starting transient state), analyzes the simulated runs to compute (a) the empirical probability distribution of final absorption states (i.e., the proportion of runs ending up in each distinct recurrent state or absorbing state) and (b) the empirical mean exit time (expected number of steps) required to leave the transient set. The program should validate these empirical findings against exact theoretical solutions derived from matrix computations.

Setup and Notation (following Durrett)

Let $(X_n)_{n\geq 0}$ be a discrete-time Markov chain on a finite state space S with transition probabilities $p(x, y) = P(X_{n+1} = y \mid X_n = x)$.

For a fixed initial state, we write

$$P_x(A) = P(A \mid X_0 = x), \qquad E_x[\,\cdot\,]$$

for the corresponding probability and expectation.

For $y \in S$, the **first hitting time of** y is

$$T_y = \min\{\, n \geq 1 : X_n = y \,\}.$$

The probability that the chain, started at y, ever returns to y is

$$\rho_{yy} = P_y(T_y < \infty).$$

Transience and Recurrence

Definition 1

A state y is **recurrent** if $\rho_{yy} = 1$ (the chain returns to y infinitely often, almost surely), and **transient** if $\rho_{yy} < 1$ (the chain visits y only finitely many times, a.s.).

Writing $x \to y$ when $\rho_{xy} = P_x(T_y < \infty) > 0$ ("x can reach y"), one shows:

- if $\rho_{xy} > 0$ but $\rho_{yx} < 1$, then x is transient;
- if x is recurrent and $x \to y$, then y is recurrent, and in fact $\rho_{yx} = 1$.

A set C is **closed** if $x \in C, y \notin C \Rightarrow p(x, y) = 0$, and **irreducible** if every pair of states in C communicates.

- A closed finite set has atleast one recurrent state.

These prove that if C is a finite closed irreducible set, then all states in C are recurrent.

Decomposition of a Finite State Space

Theorem 2 (State-space decomposition)

If S is finite, it splits as a disjoint union

$$S = T \cup R_1 \cup R_2 \cup \cdots \cup R_k,$$

where T is the set of transient states and each R_i is a closed, irreducible set of recurrent states.

Idea of proof. Take $T = \{x : \exists\, y \text{ with } x \to y \text{ but } y \not\to x\}$; every such x is transient. For any remaining state x, the set $C_x = \{y : x \to y\}$ is closed and irreducible, hence (by the recurrence lemma above) consists entirely of recurrent states; repeating this for the states not yet classified produces the recurrent classes $R_1, \ldots, R_k$.

This decomposition is exactly what lets us separate the chain into a *transient part* T that is eventually left forever, and a *recurrent part* $R = R_1 \cup \cdots \cup R_k$ that, once entered, is never left.

Block Form of P and the Fundamental Matrix

Order the states so transient states $T = \{1, \ldots, t\}$ come first and recurrent states $R = \{t+1, \ldots, t+r\}$ come second. Since R is closed, P has the block structure

$$P = \begin{pmatrix} Q & S \\ 0 & * \end{pmatrix}$$

- Q: transitions among transient states ($T \to T$).
- S: transitions from transient to recurrent states ($T \to R$).
- 0: recurrent states never return to T.

Why is $I - Q$ invertible?

Since every state in T is transient and the state space is finite, the chain eventually leaves T almost surely. Hence,

$Q^n \longrightarrow 0 \quad \Longrightarrow \quad \rho(Q) < 1$*(proof here)*, where $\rho(Q) = \max_{\lambda \in \sigma(Q)} |\lambda|$ is the spectral radius of Q.

Consequently, 1 is not an eigenvalue of Q, and therefore

$$\boxed{I - Q \text{ is invertible.}}$$

The Fundamental Matrix

Since $\rho(Q) < 1$, the matrix geometric series converges:

$$\boxed{N = (I - Q)^{-1} = I + Q + Q^2 + Q^3 + \cdots}$$

Probabilistic interpretation

Recall that $(Q^n)_{ij}$ is the probability of being at transient state j after n steps, starting from transient state i, without having left T in between. Therefore,

$$N_{ij} = \sum_{n=0}^{\infty} (Q^n)_{ij}$$

represents the expected total number of visits to transient state j, starting from transient state i, before entering R.

Thus, N is called the **fundamental matrix**.

Exit Time: Definition

Definition 3

The **exit time** from T is

$$\tau \;=\; \min\{\, n \geq 0 : X_n \in R \,\}.$$

For $i \in T$, let $t_i = E_i[\tau]$ be the expected number of steps taken to leave T when starting at i.

Exit Time: First-Step Derivation

Fix $i \in T$. Condition on the first jump X_1:

$$t_i = E_i[\tau] = 1 + \sum_{k \in T} P(X_1 = k \mid X_0 = i)\, E_k[\tau \mid X_1 = k],$$

and $E_k[\tau \mid X_1 = k] = 0$ once $k \in R$ (the chain has already exited), while it equals t_k if $k \in T$ (strong Markov property, chain restarts at k). Hence

$$t_i = 1 + \sum_{k \in T} p(i,k)\, t_k = 1 + \sum_{k \in T} Q_{ik}\, t_k, \qquad i \in T.$$

In vector form, with $\mathbf{1}$ the all-ones vector,

$$\mathbf{t} = \mathbf{1} + Q\,\mathbf{t} \implies (I - Q)\,\mathbf{t} = \mathbf{1}.$$

Exit Time: Closed Form

Theorem 4

Let $N := (I - Q)^{-1}$ (the **fundamental matrix***). Then the vector of expected exit times is*

$$\boxed{\mathbf{t} = N\mathbf{1}}$$

Exit Distribution: Definition

Definition 5

For $i \in T$ and $j \in R$, the **exit distribution** is

$$\beta_{ij} \;=\; P_i\big(X_\tau = j\big),$$

the probability that the *first* recurrent state entered, starting from i, is exactly j.

Exit Distribution: First-Step Derivation

Fix $i \in T$, $j \in R$. Condition on X_1 exactly as before:

$$\beta_{ij} = \sum_{k \in T} P(X_1 = k \mid X_0 = i)\, P_k(X_\tau = j) + \sum_{k \in R} P(X_1 = k \mid X_0 = i)\, \mathbf{1}[k = j].$$

The first sum uses the strong Markov property: if $k \in T$, the chain restarts at k and $P_k(X_\tau = j) = \beta_{kj}$ by definition. The second sum is immediate: if the chain is already *at* j at time 1, it has exited with $X_\tau = j$; if it is at some *other* recurrent state $k \neq j$, then $X_\tau = k \neq j$, contributing 0. Hence

$$\beta_{ij} = \sum_{k \in T} Q_{ik}\, \beta_{kj} + S_{ij}, \qquad i \in T,\ j \in R.$$

Exit Distribution: Closed Form

In matrix form, with $B = (\beta_{ij})_{i \in T, j \in R}$,

$$B = QB + S \implies (I - Q)B = S.$$

We summarise as:

Theorem 6

$$\boxed{B = (I - Q)^{-1} S = NS}$$

What Is Simulated and What Is Validated

Simulated (Monte Carlo): for a chosen start $i \in T$, generate n independent trajectories of the chain, each run forward one step at a time (sampling from row i of P) until it first lands in R; record the absorbing state reached and the number of steps taken.

Computed (exact): from the same Q, S blocks, the closed-form $\mathbf{t} = N\mathbf{1}$ and $B = NS$ derived above.

Validated: empirical estimates against the exact values, using a c95% confidence interval, not a raw difference — the tolerance must scale with sample size n and the variance of exit times, which a fixed threshold cannot do.

Simulation Algorithm (per start state)

```
current = start   (repeated for n independent runs)
step = 0
while current not in R:
    step += 1
    current = sample next state from row P[current]
record: absorbed_in = current, exit_steps = step
```

Empirical estimators from n runs:

$$\hat{p}_j = \frac{\#\{\text{runs absorbed in } j\}}{n}, \qquad \bar{T}_n = \frac{1}{n}\sum_{k=1}^{n} \text{exit_steps}_k.$$

Statistical Validation: Mean Exit Time

By the CLT, with $\hat{\sigma}$ the sample standard deviation of the per-run exit times,

$$\mathrm{SE} = \frac{\hat{\sigma}}{\sqrt{n}}, \qquad \frac{\bar{T}_n - t_i}{\mathrm{SE}} \xrightarrow{d} \mathcal{N}(0,1).$$

Validation check:

$$\left| \bar{T}_n - t_i \right| \leq 1.96 \cdot \mathrm{SE} \quad \text{(theoretical value inside the empirical 95\% CI).}$$

This is a two-sided z-test at the 5% level; the tolerance widens automatically for noisier chains and tightens as n grows, which a fixed $|\text{diff}| < \varepsilon$ threshold cannot do.

Pipeline: Input File $\rightarrow$ Output File

- `input.json`: transition matrix, labels, transient/recurrent index sets, list of 5–7 starting states, number of runs n.
- For each start state: simulate n trajectories, compute $(\hat{p}_j, \bar{T}_n)$, compute $(\beta_{i\cdot}, t_i)$ from $N = (I - Q)^{-1}$.
- `output.txt`: a per-start report (empirical vs. theoretical table, 95% CI check) plus a summary table across all starts.

Running from multiple transient start states lets us check that N, t and B — computed *once* from Q, S — correctly predict every row, not just one.

Sample Result (One Start State)

Final state	Empirical	Theoretical	|Diff|
Abs (R_0)	0.4209	0.4216	0.0007
Rec (R_1)	0.2958	0.2941	0.0017
Rec (R_2)	0.2834	0.2843	0.0009

$\bar{T}_n = 1.4145$ (± 0.0046, 95% CI), $t_i = 1.4107$ $\Rightarrow$ inside CI: True.

Repeating across 5–7 start states and tabulating $\bar{T}_n$ vs. t_i side by side gives a compact, reproducible check that simulation and theory agree everywhere, not just at one starting point.

Thank You

Questions?